



UNIVERSITY OF
WATERLOO

MTE 203 – Advanced Calculus

Midterm Exam Fall 2005 – Problem 2

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The intersection of the following surfaces

$$x^2 + y^2 + z^2 = 4a^2$$

$$x^2 + (y - a)^2 = a^2$$

in the positive first quadrant (i.e. $x \geq 0, y \geq 0, z \geq 0$) produces a curve defined by the equations:

$$x = a \sin 2t$$

$$y = a + a \cos 2t$$

$$z = 2a \sin t$$

- Sketch the curve
- Determine the range of increasing t and mark the direction of the curve in your graph
- Calculate the unit tangent vector $\hat{T}$ along the curve in the direction of increasing t , at $t = 0$.
- Setup the integral from which you could find the length of the curve over the range of t obtained in part (a). Do not evaluate the integral but reduce your answer to its simplest form.

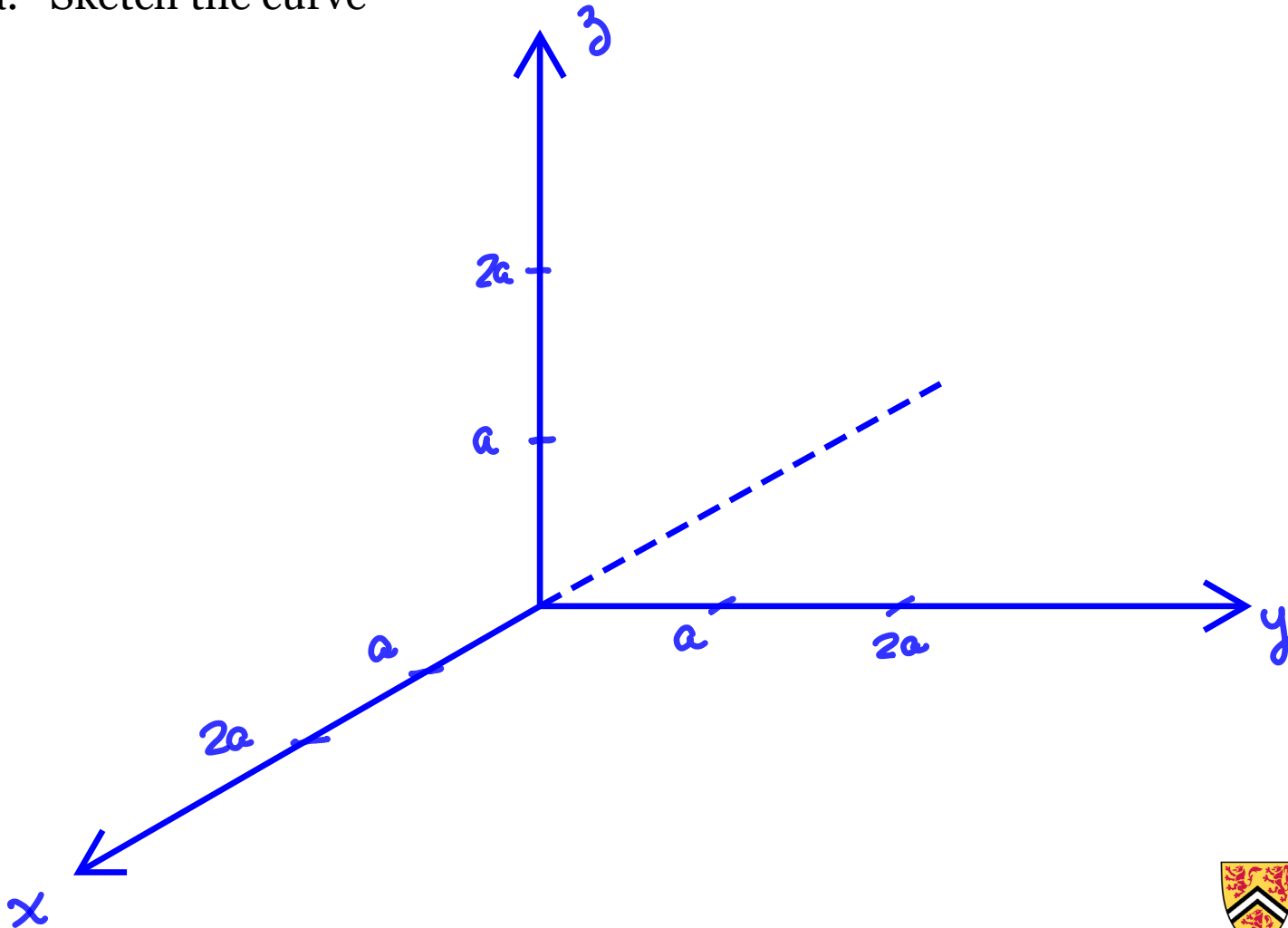
Index: Parametrization and Length of 3D Curves



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a. Sketch the curve



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- b. What is the range of t for the curve of intersection
- c. Calculate the unit tangent vector $\hat{T}$ along the curve in the direction of increasing t , at $t = 0$.

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- d. Setup the integral from which you could find the length of the curve over the range of t obtained in part (a). Do not evaluate the integral but reduce your answer to its simplest form.



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